DATABASE ARCHITECTURE & DATA MODEL SPECIFICATION

DAFCustomer Support and Monitoring System

Relational Schema, Entity Models, Foreign Keys & Entity Relationship Diagram (ERD)

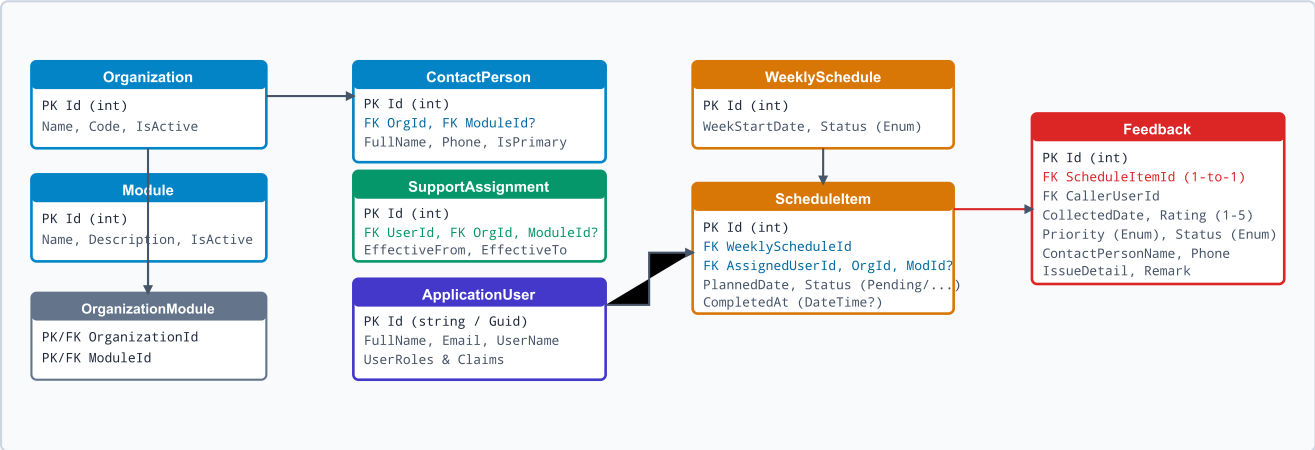
1. Architectural Overview & Data Standards

The persistence tier uses **Microsoft SQL Server** with **Entity Framework Core 9** following Code-First conventions. Standard conventions include:

- **Surrogate Keys:** Primary keys use strongly typed identifiers (`bigint` identity or `uniqueidentifier` /UUID).
- **Soft-Deletes & Auditing:** Master records implement `IsActive` flags, with automated audit tracking (`CreatedAt` , `UpdatedAt` , `CreatedBy`).
- **Immutability:** Feedback logs and completed schedule historical records are append-only.

2. Entity Relationship Diagram (ERD)

The visual relationship layout between Master Data, Operational Planning, Feedback, and Identity is presented below:



3. Complete Data Dictionary & Column Definitions

3.1 Master Data Tables

Column Name	SQL Data Type	Nullable	Constraints & Business Rules
Table: Organizations (Customers)			
PK Id	`int` (Identity)	No	Primary key surrogate identifier.
Name	`nvarchar(150)`	No	Customer organization name (Indexed).
Code	`nvarchar(20)`	No	Unique customer code (e.g., ORG-001). Unique Index.
IsActive	`bit`	No	Soft-delete status (Default = 1).
CreatedAt / UpdatedAt	`datetime2(7)`	No / Yes	UTC audit timestamps.
Table: Modules (Application Packages)			
PK Id	`int` (Identity)	No	Primary key identifier.
Name	`nvarchar(100)`	No	Software module name (e.g., Inventory, Payroll). Unique.
Description	`nvarchar(500)`	Yes	Module overview & scope description.
IsActive	`bit`	No	Status flag (Default = 1).
Table: ContactPersons (Customer Focal Points)			
PK Id	`int` (Identity)	No	Primary key.
FK OrganizationId	`int`	No	Foreign Key references `Organizations(Id)` ON DELETE CASCADE.
FK ModuleId	`int`	Yes	Optional Foreign Key references `Modules(Id)` for focal points.

Column Name	SQL Data Type	Nullable	Constraints & Business Rules
FullName	`nvarchar(100)`	No	Contact person's full name.
Phone	`nvarchar(25)`	No	Primary telephone / mobile number (FR-07).
Email	`nvarchar(100)`	Yes	Optional contact email.
IsPrimary	`bit`	No	Fallback contact if module-specific contact is absent.

3.2 Planning, Schedule & Execution Tables

Column Name	SQL Data Type	Nullable	Constraints & Business Rules
Table: WeeklySchedules			
PK Id	`int` (Identity)	No	Schedule batch identifier.
WeekStartDate / EndDate	`date`	No	Date bounds (Mon–Sun). Indexed.
Status	`int` / Enum	No	`0 = Draft`, `1 = Published` (Denominator valid only when Published).
Table: ScheduleItems (Planned Touchpoints)			
PK Id	`int` (Identity)	No	Unique scheduled contact record.
FK WeeklyScheduleId	`int`	No	References `WeeklySchedules(Id)`.
FK AssignedUserId	`nvarchar(450)`	No	References `AspNetUsers(Id)`. Filter for Agent dashboard.
FK OrganizationId / ModuleId	`int`	No / Yes	Target customer organization and optional specific module.
PlannedDate	`date`	No	Target date for call execution.
Status	`int` / Enum	No	`0 = Pending`, `1 = Completed`, `2 = Skipped`.
CompletedAt	`datetime2(7)`	Yes	Timestamp when valid feedback record was saved.
Table: Feedbacks (Collected Data & CSAT)			
PK Id	`int` (Identity)	No	Feedback entry record.
FK ScheduleItemId	`int`	No	Unique 1-to-1 link to `ScheduleItems(Id)`. Unique Index.
FK CallerUserId	`nvarchar(450)`	No	Authenticated user ID who conducted call (FR-04).
Rating	`int`	No	Check Constraint: `Rating BETWEEN 1 AND 5` (CSAT score).
Priority	`int` / Enum	No	`0 = Low, 1 = Medium, 2 = High, 3 = Blocker`.
Status	`int` / Enum	No	`0 = NotResolved, 1 = Assigned, 2 = OnProgress, 3 = Resolved`.
ContactedPersonName / Phone	`nvarchar(100) / (25)`	No	Snapshot of contacted representative details.
IssueDetail / Remark	`nvarchar(max)`	Yes / No	Detailed issue log (optional) and call remarks (mandatory).

> **Database Performance Indexing:** The query planner requires a composite index on `ScheduleItems(AssignedUserId, PlannedDate, Status)` to deliver sub-500ms response times on Agent and Lead dashboard counters.